GENCO meets the neutral wire: four-wire dynamic operating envelopes for low-voltage grids

Extending GENCO to three-phase four-wire low-voltage distribution networks: what breaks, and how a phase-aware extension repairs it.

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GridFM GENCO Challenge 2026 [1]. Code, protocols, figures and results as submitted: release 7a0325d; official-checkpoint experiments: commit 994d5d4.

Summary

GENCO models the power grid as balanced. This entry takes it to the low-voltage (LV) distribution level, where every household is connected to a single phase. To our knowledge, it is the first application of GENCO to three-phase four-wire LV networks. We test it on household dynamic operating envelopes:

- **A balanced model is unsafe at LV.** Across every hour of 36 evaluation days, not one positive envelope designed with the balanced model met the four-wire feasibility target.
- **A phase-aware extension of the GENCO architecture makes it useful.** Used as a pre-screen, it cuts full four-wire OpenDSS simulations six-fold (fallbacks 35.8% → 5.8%), with zero observed false-safe decisions on 2,968 evaluation states.
- **The official pretrained GENCO transfers when the representation matches.** Fine-tuning the official checkpoint on case118 beat training from scratch by **+11.4 points [9.8, 12.5], in 5 of 5 seeds**, on 2,000 unseen scenarios.
- **Dynamic envelopes release real headroom:** 4.81 MWh of export and 2.50 MWh of import envelope-hours per household per year, where a fixed allowance valid in every hour would be zero.
- **Toward integrated DSO–TSO tools.** Household envelopes add up at the MV/LV transformer into the flexibility a distribution system operator (DSO) can offer the transmission system operator (TSO). This entry works at both ends of that chain with one model family (four-wire LV and case118 transmission) and identifies what a single model across voltage levels needs.

1. The problem

Distribution operators are replacing fixed connection limits with **dynamic operating envelopes (DOEs)** [5]. A DOE is a time-varying allowance that tells each household how much extra power it may import (EV charging, heat pumps) or export (rooftop PV) in each interval, without the network leaving its operational limits. Summed over a feeder, the envelopes are also the flexibility the operator can offer to the medium-voltage (MV) and transmission levels. Computing them means solving many power flows under uncertainty, which is the kind of task a fast learned grid model could take over. Today DSOs and TSOs coordinate with separate tools built on separate physics; a grid foundation model covering every voltage level could be the shared layer between them. That only works if it handles LV physics, which is what we test.

GENCO [2] is such a model: a grid foundation model, available pretrained on transmission cases, that represents the network as **balanced**, with one equivalent line per bus and no neutral conductor. European LV networks are **three-phase four-wire**: each home is connected between one phase and the neutral, the phases are loaded unevenly, and the neutral carries the imbalance. One phase can drop below the voltage limit while another rises above it, which a balanced model cannot capture. We asked: **what would it take to extend GENCO to four-wire LV networks, and can it then support household envelopes?**

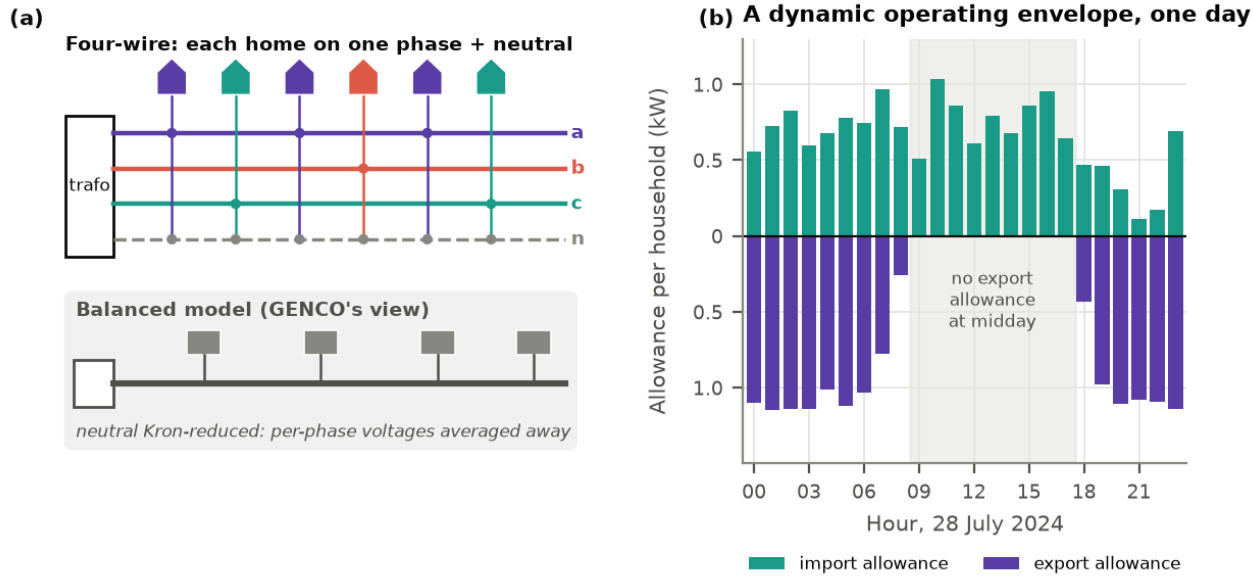


Figure 1. *(a) A four-wire LV network, and the balanced view GENCO works with: homes lumped per bus, neutral removed. (b) One day (28 July 2024) of envelopes on our feeder that passed the four-wire check, per household and hour, for import (up) and export (down).*

Definitions

- **Envelope / envelope-hours.** One allowance in kW, the same for every household, on top of its own demand, per hour and direction. Summed over a year, it gives the headroom made available, not energy actually used.
- **OpenDSS** [6]. An open-source distribution-system simulator; its full three-phase four-wire power flow is our ground truth.
- **Acceptance rule.** An envelope is issued only if OpenDSS finds the network within limits in at least 98% of 200 design scenarios for that hour (a 5% violation level at 95% confidence).

- **Screen, fallback, false-safe.** A screen labels a network state *safe*, *unsafe* or *unsure*; unsure states fall back to OpenDSS. A *false-safe* decision calls an unsafe state safe, and is the error that matters.
- **Kron reduction.** Folding the neutral into the phase conductors, the first step of building a balanced model.
- **One-step four-wire sweep.** A single iteration of the standard radial backward/forward sweep on the four-wire network: a cheap, rough physics estimate.

2. Why the balanced representation fails at LV

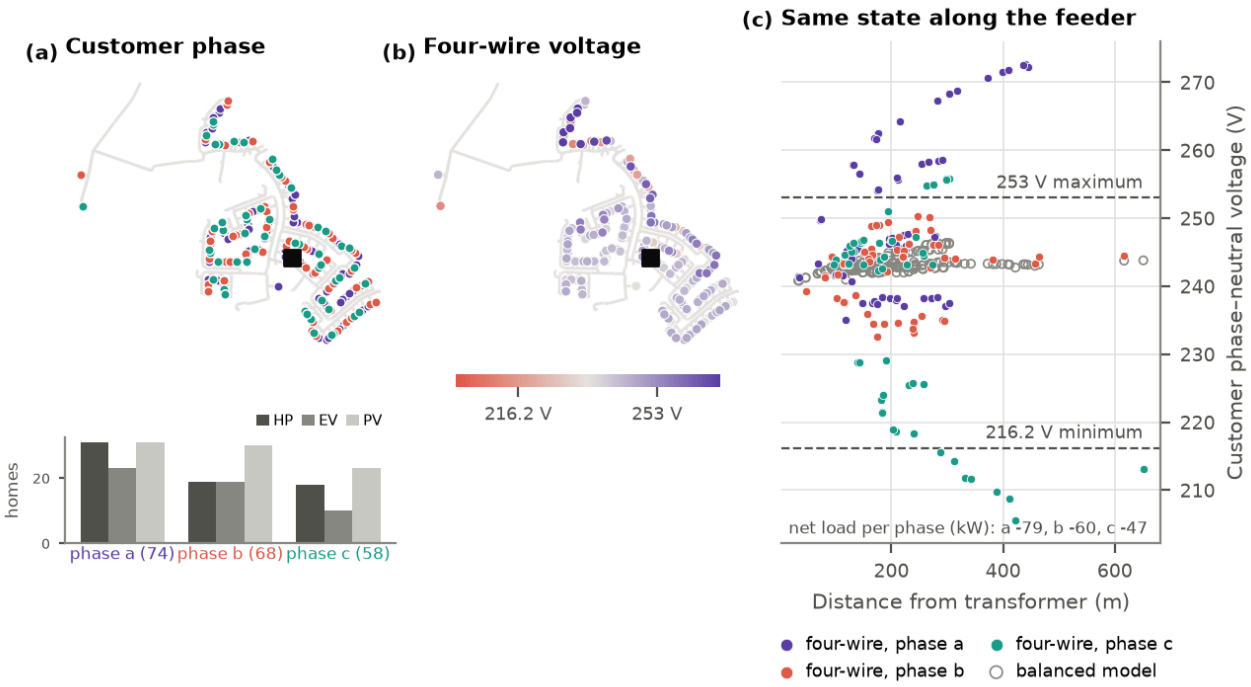


Figure 2. *ENWL feeder 1. (a) Connection phase of each customer, and the heat pumps (HP), EVs and PV per phase in the headline household placement. (b, c) One validation state with 186 kW net export (–79, –60 and –47 kW on phases a, b and c). The balanced model puts every customer between 240 and 247 V, inside the 216.2–253.0 V limits. The four-wire solution of the same injections ranges from 205 V on phase c to above 270 V on phase a: both limits are violated at once.*

A balanced model cannot capture voltage deviations on individual phases, such as a rise on one and a drop on another. That unbalance drives elevated neutral currents and shifts the neutral point. For envelopes the consequence is decisive: across every hour of the 36 evaluation days, not one positive balanced-model envelope met the four-wire feasibility target (0 of 864 export hours, 0 of 823 import hours).

3. Method

3.1 Making the GENCO architecture work on an LV feeder

We trained the GENCO architecture in gridfm-graphkit [3] (`GNS_heterogeneous` as distributed, the *stock* model) on this feeder. It did worse than a trivial baseline that predicts each bus's average training-set voltage. We traced this to **three bottlenecks of GENCO at LV**:

1. **The edge encoder cannot see impedance magnitude.** It ends in a layer normalisation that, with LV admittances of order 1e4, strongly suppresses the magnitude that distinguishes a long cable from a short one. Multiplying every admittance by 10 changes the edge embedding by only 0.2%. *Fix:* asinh-scaled admittance features.
2. **The physics loss collapses at LV numerical scales.** *Fix:* a residual head that predicts a correction to a linear power-flow estimate, initialised at zero, so training starts exactly at the physics estimate.
3. **Line flows are ill-conditioned.** Flows from voltage differences turn a 1e-4 pu voltage error into an O(100%) flow error on short cables. *Fix:* flows from Kirchhoff's current law.

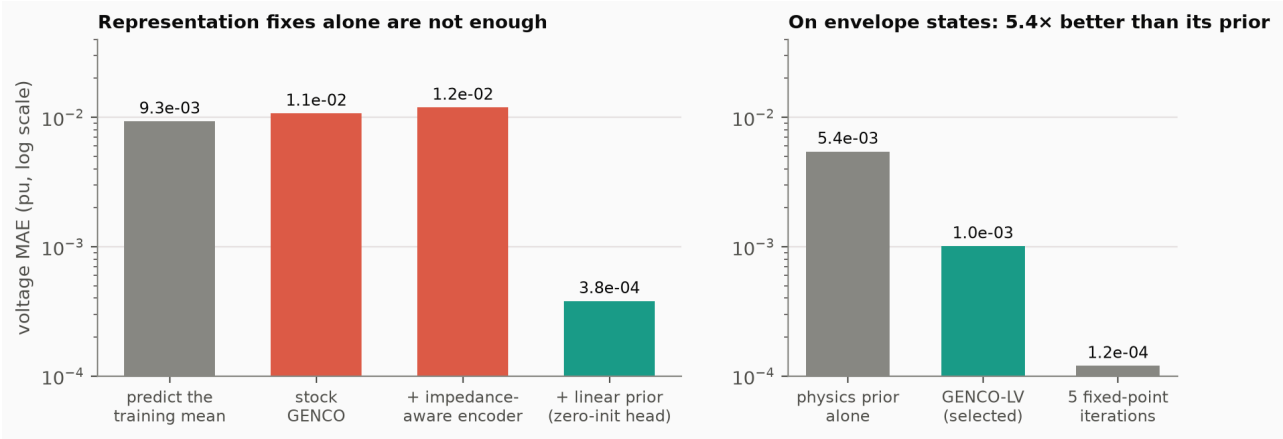


Figure 3. *Voltage error (mean absolute error, pu, log scale) of each modelling step. Left: 1,000 validation states, 15-epoch screening runs. Right: 1,000 validation states including envelope-loaded conditions, final runs. Separate experiments, not one ablation sequence. Coral: fails; grey: baselines.*

The admittance features were necessary but not sufficient. The decisive step was combining them with the physics estimate and residual head: on envelope-loaded states the error fell to 1.0e-3 pu, 5.4× below the estimate alone.

We also found and fixed two bugs in the official GridFM tooling before any reported run. graphkit's best-model export could save the untrained epoch-0 weights, so we checked every reported export, including the stock model, against its last checkpoint. datakit's optimal-power-flow step raises an unpinned slack bus to its upper voltage bound (1.5 pu with our loose bounds), so we fixed the slack at 1.0 pu.

3.2 Making it phase-aware

What decides feasibility here is the voltage on each phase and the neutral-point displacement, which no balanced representation can carry, however long it is trained. We therefore extended the GENCO architecture with a full three-phase four-wire network model: phase and neutral conductors, each line described by its 4×4 impedance matrix, and no Kron reduction. Its attention-based message passing (TransformerConv) carries those impedances. As in 3.1, it predicts a correction to a cheap physics estimate, here the one-step four-wire sweep.

3.3 GENCO pre-screening to reduce OpenDSS simulations

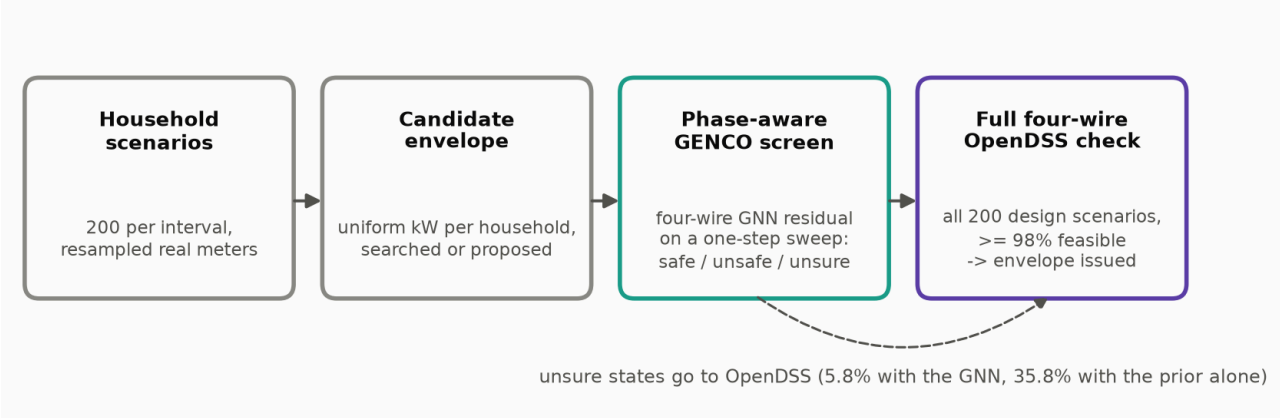


Figure 4. *The phase-aware GENCO screen decides clear cases. Unsure states, and every envelope before it is issued, go through the full four-wire OpenDSS check.*

The final check is necessary: without it, a GENCO screen let one materially unsafe envelope through in 1,728 intervals. It verifies each envelope on its design scenarios, not on all possible futures. On new scenarios, issued non-zero envelopes average 97.4% four-wire feasibility, but only 47% pass a one-sided 95% lower-bound test. Proving that would need more draws, each adding OpenDSS simulations for every candidate envelope; 200 was the pre-registered design size.

4. Experimental setup

- **Network.** The open-source benchmark ENWL1 (three-phase four-wire model, UK): 800 kVA, 200 single-phase customers, 4×4 cable impedances from BS 5467 data. It is modified to TT earthing, with the neutral earthed only at the transformer, an important choice that sets the size of the neutral shift. The balanced model Kron-reduces the neutral, takes each line's positive-sequence impedance, and lumps each customer as a three-phase load at its bus.
- **Households.** Fluvius open data: 2,400 anonymised residential digital meters, quarter-hourly, 2024, in eight technology groups [4]. Technology mix: 30% heat pumps, 30% EVs, 40% rooftop PV.
- **Limits and checks.** Voltage within –6%/+10% of 230 V, i.e. 216.2–253.0 V (UK ESQCR [7]). Export envelopes are checked for overvoltage and for phase-conductor and transformer loading; import envelopes for undervoltage and the same loadings.
- **Scenarios.** The 200 design scenarios of an hour resample real meters of the same technology group at the *same date and time*, so the spread reflects household diversity at that moment, not seasonal variation.
- **Evaluation.** Envelopes on 36 days spaced ten days apart through 2024 (10 January to 25 December), hourly, both directions. Screens on 2,968 four-wire states from those days, including envelope-loaded states, all under the same final check. Learned models were trained on separate 2024 days. Baselines: a balanced sweep, a balanced GENCO screen, and one-step and iterated four-wire sweeps.
- **Evidence status.** The full OpenDSS envelope search and the GENCO screen were pre-registered. Everything else is exploratory, with protocols frozen before results. The evalua-

tion days were also used in development. Each learned model uses one seed unless stated otherwise.

5. Results

5.1 Screening performance

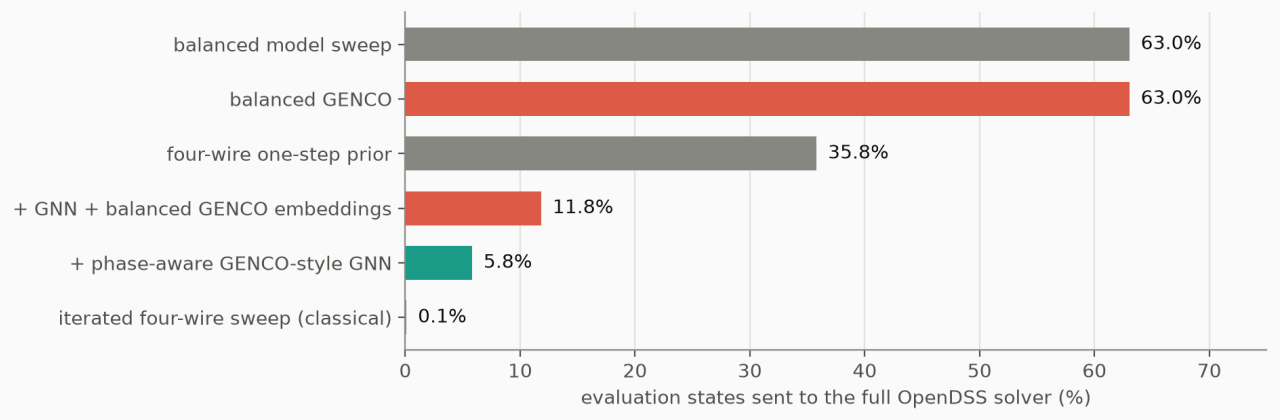


Figure 5. *Share of the 2,968 evaluation states each screen sends to OpenDSS, all under the same final check; none made a false-safe decision. Teal: the phase-aware GENCO extension; coral: balanced-GENCO variants; grey: physics baselines.*

- **Fallbacks fall from 35.8% to 5.8%** relative to the one-step four-wire sweep, with **zero observed false-safe decisions**: six-fold fewer OpenDSS simulations.
- Phase-to-neutral voltage error falls by 91% (on the 2,699 converged states), and near-limit violation classification improves from 0.926 to 0.992 balanced accuracy.
- Balanced screens, a balanced sweep or balanced GENCO, send 63% of states to OpenDSS. **Representing the phases is what makes screening useful.**
- Adding embeddings from our feeder-specific balanced GENCO *raised* fallbacks to 11.8% (+6.0 points [5.3, 6.7]). We did not isolate why. A plausible reading is that the embeddings describe a balanced state that disagrees with the four-wire one exactly where screening matters.
- Per state, the screen takes 2.1 ms on an A100 GPU against 5.5 ms for our CPU four-wire sweep. The hardware differs, so the main computational gain is the six-fold reduction in OpenDSS simulations.

5.2 The official pretrained GENCO

On case118 with 99 flexible export units, a network proposes the flexible dispatch and an AC optimal power flow (PowerModels [9] with IPOPT [8]) re-dispatches the rest. The reference is the IPOPT optimum. Fine-tuning the official `gridfm/genco-opf-datakit` checkpoint beat training from scratch at 100 labels by **+11.4 points of reference hosting [9.8, 12.5], in 5 of 5 seeds**, on 2,000 unseen scenarios. The gain also held at 10,000 labels in every seed; at 1,000 labels it depended on the seed. Our pre-registered absolute target of 95% of the reference was narrowly missed (94.7%). (log) Pretraining transfers when the representation matches, but it cannot supply missing phases.

5.3 Operational value

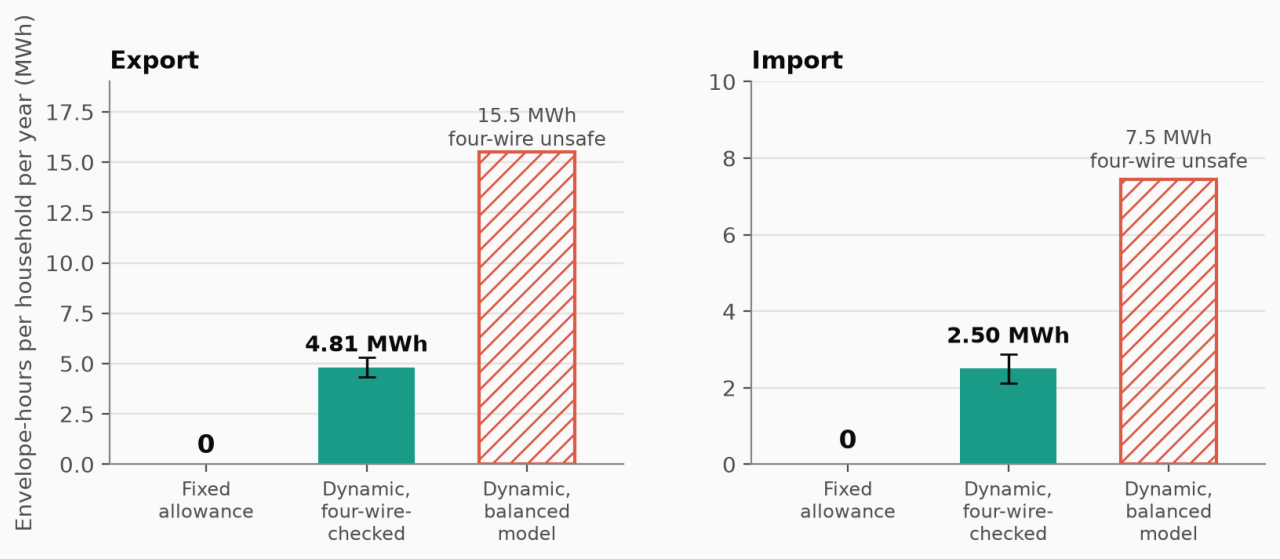


Figure 6. *Envelope-hours per household per year over the 36 evaluation days: a fixed allowance valid in every hour, dynamic envelopes that passed the four-wire check (with season-stratified day-bootstrap 95% intervals), and balanced-model envelopes, which fail it.*

Dynamic envelopes release **4.81 MWh of export** (4.35–5.30) and **2.50 MWh of import** (2.11–2.87) envelope-hours per household per year. A fixed allowance valid in every hour would be zero: in about half of all hours no positive allowance passes the check. Across five household-technology placements the result is 4.8–6.2 MWh export and 2.5–3.2 MWh import; the headline placement is the lowest. This is headroom an operator could offer for EV charging, heat pumps or PV export on the existing feeder; we do not estimate its uptake or its value for reinforcement deferral. These figures come from the full OpenDSS search. Running the phase-aware screen through the full-year study is the next step.

6. Conclusions and outlook

1. **Balanced grid models, GENCO included, are not safe proxies for household envelopes on four-wire LV networks.**
2. **The GENCO architecture can be extended to the distribution level** with magnitude-preserving edge features, a physics residual and an explicit phase-and-neutral network model. As a pre-screen it cuts full four-wire simulations six-fold without an observed false-safe decision.
3. **Pretraining transfers when the representation matches** (case118, 5 of 5 seeds), but it cannot supply the missing phases.
4. **On this feeder today, physics stays in charge.** The iterated four-wire sweep remains the accuracy leader (3.6e-6 pu, 0.1% fallback, about 80× more accurate). The learned screen's value is fewer OpenDSS simulations, and amortisation across networks and workflows where an equally strong specialised solver is not available.

Limitations. The check assumes every household uses its full envelope (partial uptake can shift the neutral further). Each direction is checked separately, so export hours are not checked for un-

dervoltage on other phases. Neutral-conductor current and voltage unbalance are not checked. The study covers one benchmark feeder.

Outlook: toward integrated DSO–TSO tools. The next step is to aggregate the household envelopes into a P/Q flexibility range per MV/LV transformer, feed it into an MV stage (exploratory MV work on case33bw is in the repository), and pass the MV aggregate to the transmission level. One model across these levels needs what this entry identified: magnitude-preserving edge features, level-specific representations (phase-aware at LV, positive-sequence at HV), and a physics check at every level.

7. Reproducibility

- **Run it yourself, with public data only.** `python scripts/demo_neutral_wire.py` solves one deliberately stressed synthetic state on the public ENWL1 files with both models in a few seconds. The balanced model shows no customer outside the limits. Four-wire OpenDSS shows 34, from 205 to 257 V, with 401 A in the transformer neutral and currents above cable ratings on some segments.
- **Code, protocols and results** are at the release commit (Apache-2.0): numbered amendments, the experiment log, run manifests with SHA-256 hashes, claims and non-claims, and 84 tests. `scripts/reproduce_check.py` regenerates every headline number byte for byte.
- **Public inputs.** The ENWL1 files are in the repository and the household profiles are Fluvius open data [4], so every result can be regenerated from public inputs. We have not yet redistributed the derived files (LV checkpoints, four-wire labels, per-day run files) while we check the licence terms for meter-level derivatives; their hashes are in the manifests.

Built on GENCO with gridfm-graphkit 0.9.0.3 and gridfm-datakit 1.1.0 [3].

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